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| <b>Lab Task:</b> | 09 |
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| <b>Course Code:</b> | CL2005 |
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| <b>Course Title:</b> | Database Systems - Lab |
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| <b>Semester:</b> | Spring 2026 |
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| <b>Instructor:</b> | Muhammad Mehdi |
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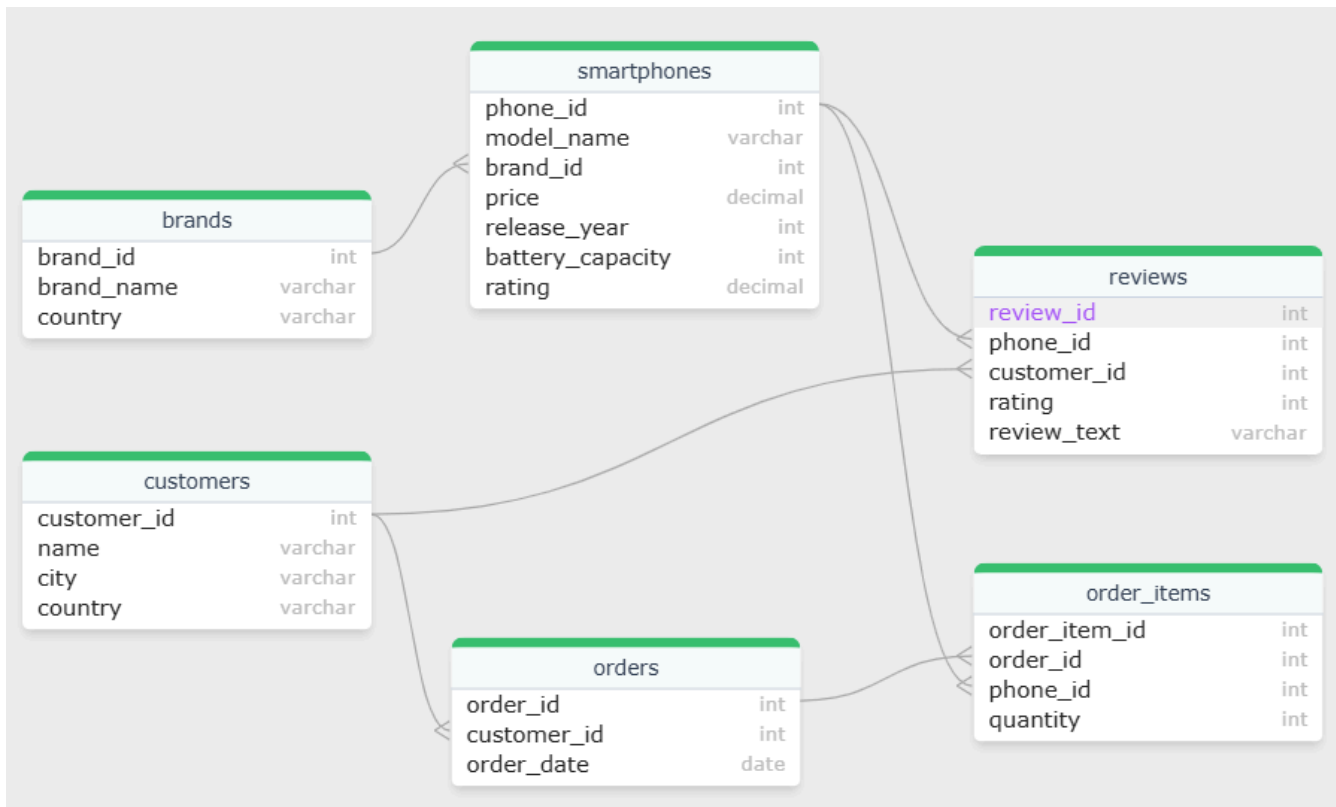
## Guidelines

- Please review the lab manual before starting the task to ensure you fully understand the requirements.
- You may use **only the concepts covered in the lab manual** to complete this task.
- Your submission must be a **single PDF file** that includes:
  - The source code (in plain text)
  - Screenshots of the corresponding outputs
- Name your file using the following format: **rollno\_name\_labtasknumber.pdf**  
**Example:** *24p-1234\_ozair\_labtask09.pdf*
- Unethical use of AI tools will result in a **deduction of 5 marks**.
- Late submissions will incur a penalty of **-1 mark per day**.

## Tasks

You are provided with a SQL script file named *lab-09-smartphones.sql*, which contains SQL statements to create and populate the required tables using a simplified, dataset based on smartphone brands, models, customers, and orders.

The complete schema of the database, consisting of all tables and the relationships among them is described in the given ER Diagram:



Execute the given script file in MySQL server using the **source** command to load the dataset. After successfully setting up the database, perform all the following tasks using subqueries:

1. Display all smartphones with the second highest rating.
2. Show smartphones that belong to brands from the USA.
3. List all customers who have placed at least one order using the **IN** operator.
4. Find the cheapest smartphone for each brand using a correlated subquery.
5. Find customers who have not written any reviews using **NOT EXISTS**.
6. Display smartphones whose battery capacity is greater than all smartphones released before 2022.
7. Display each smartphone along with the average rating of its brand using a subquery in the **SELECT** clause.



8. Display each customer name and the total number of orders placed by each customer.
9. Update the price of smartphones by decreasing 20% for those whose rating is below the average rating.
10. Delete record of smartphones that have never been ordered.